nndl/softmax.py

```python
import numpy as np


class Softmax(object):

  def __init__(self, dims=[10, 3073]):
    self.init_weights(dims=dims)

  def init_weights(self, dims):
    """
    Initializes the weight matrix of the Softmax classifier.
    Note that it has shape (C, D) where C is the number of
    classes and D is the feature size.
    """
    self.W = np.random.normal(size=dims) * 0.0001

  def loss(self, X, y):
    """
    Calculates the softmax loss.

    Inputs have dimension D, there are C classes, and we operate on minibatches
    of N examples.

    Inputs:
    - X: A numpy array of shape (N, D) containing a minibatch of data.
    - y: A numpy array of shape (N,) containing training labels; y[i] = c means
      that X[i] has label c, where 0 ≤ c < C.

    Returns a tuple of:
    - loss as single float
    """

    # Initialize the loss to zero.
    loss = 0.0

    # ================================================================ #
    # YOUR CODE HERE:
    #   Calculate the normalized softmax loss.  Store it as the variable loss.
    #   (That is, calculate the sum of the losses of all the training
    #   set margins, and then normalize the loss by the number of
    #   training examples.)
    # ================================================================ #

    num_train = X.shape[0]
    scores = X.dot(self.W.T)

    # Numerical stability
    scores -= np.max(scores, axis=1, keepdims=True)

    # Softmax probabilities
    probs = np.exp(scores) / np.sum(np.exp(scores), axis=1, keepdims=True)
```

```python

    # Compute the loss: average cross-entropy loss
    correct_logprobs = -np.log(probs[np.arange(num_train), y])
    loss = np.sum(correct_logprobs) / num_train

    # ================================================================ #
    # END YOUR CODE HERE
    # ================================================================ #

    return loss

  def loss_and_grad(self, X, y):
    """
    Same as self.loss(X, y), except that it also returns the gradient.

    Output: grad -- a matrix of the same dimensions as W containing
      the gradient of the loss with respect to W.
    """

    # Initialize the loss and gradient to zero.
    loss = 0.0
    grad = np.zeros_like(self.W)

    # ================================================================ #
    # YOUR CODE HERE:
    #   Calculate the softmax loss and the gradient. Store the gradient
    #   as the variable grad.
    # ================================================================ #

    num_train = X.shape[0]
    num_classes = self.W.shape[0]

    for i in range(num_train):
        # 1. Compute score for the i-th example
        scores = X[i].dot(self.W.T)

        # 2. Stability trick (shift by max)
        scores -= np.max(scores)

        # 3. Softmax probability
        exp_scores = np.exp(scores)
        sum_exp_scores = np.sum(exp_scores)
        probs = exp_scores / sum_exp_scores

        # 4. Accumulate Loss
        loss += -np.log(probs[y[i]])

        # 5. Accumulate Gradient
        # dL/dW_c = (p_c - 1) * x_i  if c == y_i
        # dL/dW_c = p_c * x_i        if c != y_i
        for c in range(num_classes):
            p = probs[c]
            if c == y[i]:
                grad[c, :] += (p - 1) * X[i]
```

```
            else:
                grad[c, :] += p * X[i]

    # Average over the batch
    loss /= num_train
    grad /= num_train

    # ================================================================ #
    # END YOUR CODE HERE
    # ================================================================ #

    return loss, grad

  def grad_check_sparse(self, X, y, your_grad, num_checks=10, h=1e-5):
    """
    sample a few random elements and only return numerical
    in these dimensions.
    """

    for i in np.arange(num_checks):
      ix = tuple([np.random.randint(m) for m in self.W.shape])

      oldval = self.W[ix]
      self.W[ix] = oldval + h # increment by h
      fxph = self.loss(X, y)
      self.W[ix] = oldval - h # decrement by h
      fxmh = self.loss(X,y) # evaluate f(x - h)
      self.W[ix] = oldval # reset

      grad_numerical = (fxph - fxmh) / (2 * h)
      grad_analytic = your_grad[ix]
      rel_error = abs(grad_numerical - grad_analytic) / (abs(grad_numerical) + abs(grad_analytic))
      print('numerical: %f analytic: %f, relative error: %e' % (grad_numerical, grad_analytic, rel_error))

  def fast_loss_and_grad(self, X, y):
    """
    A vectorized implementation of loss_and_grad. It shares the same
    inputs and ouptuts as loss_and_grad.
    """
    loss = 0.0
    grad = np.zeros(self.W.shape) # initialize the gradient as zero

    # ================================================================ #
    # YOUR CODE HERE:
    #   Calculate the softmax loss and gradient WITHOUT any for loops.
    # ================================================================ #

    num_train = X.shape[0]
    scores = X.dot(self.W.T)

    scores -= np.max(scores, axis=1, keepdims=True)

```

```
    probs = np.exp(scores) / np.sum(np.exp(scores), axis=1, keepdims=True)

    loss = np.sum(-np.log(probs[np.arange(num_train), y])) / num_train

    dscores = probs
    dscores[np.arange(num_train), y] -= 1
    grad = dscores.T.dot(X) / num_train

    # ================================================================ #
    # END YOUR CODE HERE
    # ================================================================ #

    return loss, grad

  def train(self, X, y, learning_rate=1e-3, num_iters=100,
            batch_size=200, verbose=False):
    """
    Train this linear classifier using stochastic gradient descent.

    Inputs:
    - X: A numpy array of shape (N, D) containing training data; there are N
      training samples each of dimension D.
    - y: A numpy array of shape (N,) containing training labels; y[i] = c
      means that X[i] has label 0 ≤ c < C for C classes.
    - learning_rate: (float) learning rate for optimization.
    - num_iters: (integer) number of steps to take when optimizing
    - batch_size: (integer) number of training examples to use at each step.
    - verbose: (boolean) If true, print progress during optimization.

    Outputs:
    A list containing the value of the loss function at each training iteration.
    """
    num_train, dim = X.shape
    num_classes = np.max(y) + 1 # assume y takes values 0...K-1 where K is number of classes

    self.init_weights(dims=[np.max(y) + 1, X.shape[1]]) # initializes the weights of self.W

    # Run stochastic gradient descent to optimize W
    loss_history = []

    for it in np.arange(num_iters):
      X_batch = None
      y_batch = None

      # ================================================================ #
      # YOUR CODE HERE:
      #   Sample batch_size elements from the training data for use in
      #     gradient descent.  After sampling,
      #     - X_batch should have shape: (batch_size, dim)
      #     - y_batch should have shape: (batch_size,)
      #   The indices should be randomly generated to reduce correlations
      #   in the dataset.  Use np.random.choice.  It's okay to sample with
```

```
      #   replacement.
      # ================================================================ #

      indices = np.random.choice(num_train, batch_size)
      X_batch = X[indices]
      y_batch = y[indices]

      # ================================================================ #
      # END YOUR CODE HERE
      # ================================================================ #

      # evaluate loss and gradient
      loss, grad = self.fast_loss_and_grad(X_batch, y_batch)
      loss_history.append(loss)

      # ================================================================ #
      # YOUR CODE HERE:
      #   Update the parameters, self.W, with a gradient step
      # ================================================================ #

      self.W -= learning_rate * grad

      # ================================================================ #
      # END YOUR CODE HERE
      # ================================================================ #

      if verbose and it % 100 == 0:
        print('iteration {} / {}: loss {}'.format(it, num_iters, loss))

    return loss_history

  def predict(self, X):
    """
    Inputs:
    - X: N x D array of training data. Each row is a D-dimensional point.

    Returns:
    - y_pred: Predicted labels for the data in X. y_pred is a 1-dimensional
      array of length N, and each element is an integer giving the predicted
      class.
    """
    y_pred = np.zeros(X.shape[1])
    # ================================================================ #
    # YOUR CODE HERE:
    #   Predict the labels given the training data.
    # ================================================================ #

    scores = X.dot(self.W.T)
    y_pred = np.argmax(scores, axis=1)

    # ================================================================ #
    # END YOUR CODE HERE
    # ================================================================ #

```

```
    return y_pred


```

This is the softmax workbook for ECE C147/C247 Assignment #2

Please follow the notebook linearly to implement a softmax classifier.

Please print out the workbook entirely when completed.

The goal of this workbook is to give you experience with training a softmax classifier.

In [1]:
```
import random
import numpy as np
from utils.data_utils import load_CIFAR10
import matplotlib.pyplot as plt

%matplotlib inline
%load_ext autoreload
%autoreload 2
```

In [2]:
```
def get_CIFAR10_data(num_training=49000, num_validation=1000, num_test=10
    """
    Load the CIFAR-10 dataset from disk and perform preprocessing to prep
    it for the linear classifier. These are the same steps as we used for
    SVM, but condensed to a single function.
    """
    # Load the raw CIFAR-10 data
    cifar10_dir = 'cifar-10-batches-py' # You need to update this line
    X_train, y_train, X_test, y_test = load_CIFAR10(cifar10_dir)

    # subsample the data
    mask = list(range(num_training, num_training + num_validation))
    X_val = X_train[mask]
    y_val = y_train[mask]
    mask = list(range(num_training))
    X_train = X_train[mask]
    y_train = y_train[mask]
    mask = list(range(num_test))
    X_test = X_test[mask]
    y_test = y_test[mask]
    mask = np.random.choice(num_training, num_dev, replace=False)
    X_dev = X_train[mask]
    y_dev = y_train[mask]

    # Preprocessing: reshape the image data into rows
    X_train = np.reshape(X_train, (X_train.shape[0], -1))
    X_val = np.reshape(X_val, (X_val.shape[0], -1))
    X_test = np.reshape(X_test, (X_test.shape[0], -1))
    X_dev = np.reshape(X_dev, (X_dev.shape[0], -1))

    # Normalize the data: subtract the mean image
    mean_image = np.mean(X_train, axis = 0)
    X_train -= mean_image
    X_val -= mean_image
    X_test -= mean_image
    X_dev -= mean_image
```

```
    # add bias dimension and transform into columns
    X_train = np.hstack([X_train, np.ones((X_train.shape[0], 1))])
    X_val = np.hstack([X_val, np.ones((X_val.shape[0], 1))])
    X_test = np.hstack([X_test, np.ones((X_test.shape[0], 1))])
    X_dev = np.hstack([X_dev, np.ones((X_dev.shape[0], 1))])

    return X_train, y_train, X_val, y_val, X_test, y_test, X_dev, y_dev


# Invoke the above function to get our data.
X_train, y_train, X_val, y_val, X_test, y_test, X_dev, y_dev = get_CIFAR1
print('Train data shape: ', X_train.shape)
print('Train labels shape: ', y_train.shape)
print('Validation data shape: ', X_val.shape)
print('Validation labels shape: ', y_val.shape)
print('Test data shape: ', X_test.shape)
print('Test labels shape: ', y_test.shape)
print('dev data shape: ', X_dev.shape)
print('dev labels shape: ', y_dev.shape)
```

```
Train data shape:  (49000, 3073)
Train labels shape:  (49000,)
Validation data shape:  (1000, 3073)
Validation labels shape:  (1000,)
Test data shape:  (1000, 3073)
Test labels shape:  (1000,)
dev data shape:  (500, 3073)
dev labels shape:  (500,)
```

Training a softmax classifier.

The following cells will take you through building a softmax classifier. You will implement its loss function, then subsequently train it with gradient descent. Finally, you will choose the learning rate of gradient descent to optimize its classification performance.

In [3]:
```
from nndl import Softmax
```

In [4]:
```
# Declare an instance of the Softmax class.
# Weights are initialized to a random value.
# Note, to keep people's first solutions consistent, we are going to use

np.random.seed(1)

num_classes = len(np.unique(y_train))
num_features = X_train.shape[1]

softmax = Softmax(dims=[num_classes, num_features])
```

Softmax loss

In [5]:
```
## Implement the loss function of the softmax using a for loop over
#  the number of examples

loss = softmax.loss(X_train, y_train)
```

In [6]:
```
print(loss)
```

```
2.3277607028048757
```

Question:

You'll notice the loss returned by the softmax is about 2.3 (if implemented correctly). Why does this make sense?

Hint: Connect to probability.

Answer:

At initialization, the weights are set to small random values, resulting in scores close to zero. Consequently, the exponential of these scores is approximately 1. Since there are 10 classes in the CIFAR-10 dataset, the normalization term (sum of exponentials) is approximately 10. This leads to a probability distribution where each class has a probability of roughly $1/10$ or 0.1. The cross-entropy loss is calculated as $-\ln(probability)$, so $-\ln(0.1) = \ln(10) \approx 2.3$.

Softmax gradient

In [7]:
```
## Calculate the gradient of the softmax loss in the Softmax class.
# For convenience, we'll write one function that computes the loss
#   and gradient together, softmax.loss_and_grad(X, y)
# You may copy and paste your loss code from softmax.loss() here, and the
#   use the appropriate intermediate values to calculate the gradient.

loss, grad = softmax.loss_and_grad(X_dev,y_dev)

# Compare your gradient to a gradient check we wrote.
# You should see relative gradient errors on the order of 1e-07 or less i
softmax.grad_check_sparse(X_dev, y_dev, grad)
```

```
numerical: -0.060115 analytic: -0.060115, relative error: 4.641954e-07
numerical: 0.426604 analytic: 0.426604, relative error: 2.760104e-08
numerical: -0.321928 analytic: -0.321928, relative error: 1.605465e-08
numerical: 1.928266 analytic: 1.928266, relative error: 2.682734e-10
numerical: 1.739934 analytic: 1.739934, relative error: 2.684533e-08
numerical: 2.153248 analytic: 2.153248, relative error: 4.933154e-09
numerical: 0.529605 analytic: 0.529605, relative error: 4.501774e-08
numerical: -0.738285 analytic: -0.738285, relative error: 4.895991e-09
numerical: 1.596614 analytic: 1.596614, relative error: 6.239973e-09
numerical: -3.138518 analytic: -3.138519, relative error: 1.742244e-08
numerical: -3.138518 analytic: -3.138519, relative error: 1.742244e-08
```

A vectorized version of Softmax

To speed things up, we will vectorize the loss and gradient calculations. This will be helpful for stochastic gradient descent.

In [8]:
```
import time
```

In [9]:
```
## Implement softmax.fast_loss_and_grad which calculates the loss and gra
#    WITHOUT using any for loops.

# Standard loss and gradient
tic = time.time()
loss, grad = softmax.loss_and_grad(X_dev, y_dev)
toc = time.time()
print('Normal loss / grad_norm: {} / {} computed in {}s'.format(loss, np.

tic = time.time()
loss_vectorized, grad_vectorized = softmax.fast_loss_and_grad(X_dev, y_de
toc = time.time()
print('Vectorized loss / grad: {} / {} computed in {}s'.format(loss_vecto

# The losses should match but your vectorized implementation should be mu
print('difference in loss / grad: {} /{} '.format(loss - loss_vectorized,

# You should notice a speedup with the same output.
```

```
Normal loss / grad_norm: 2.3259906233898517 / 315.69370372333213 computed
in 0.03464198112487793s
Vectorized loss / grad: 2.3259906233898517 / 315.69370372333213 computed i
n 0.013659000396728516s
difference in loss / grad: 0.0 /2.090770683360497e-13
```

Stochastic gradient descent

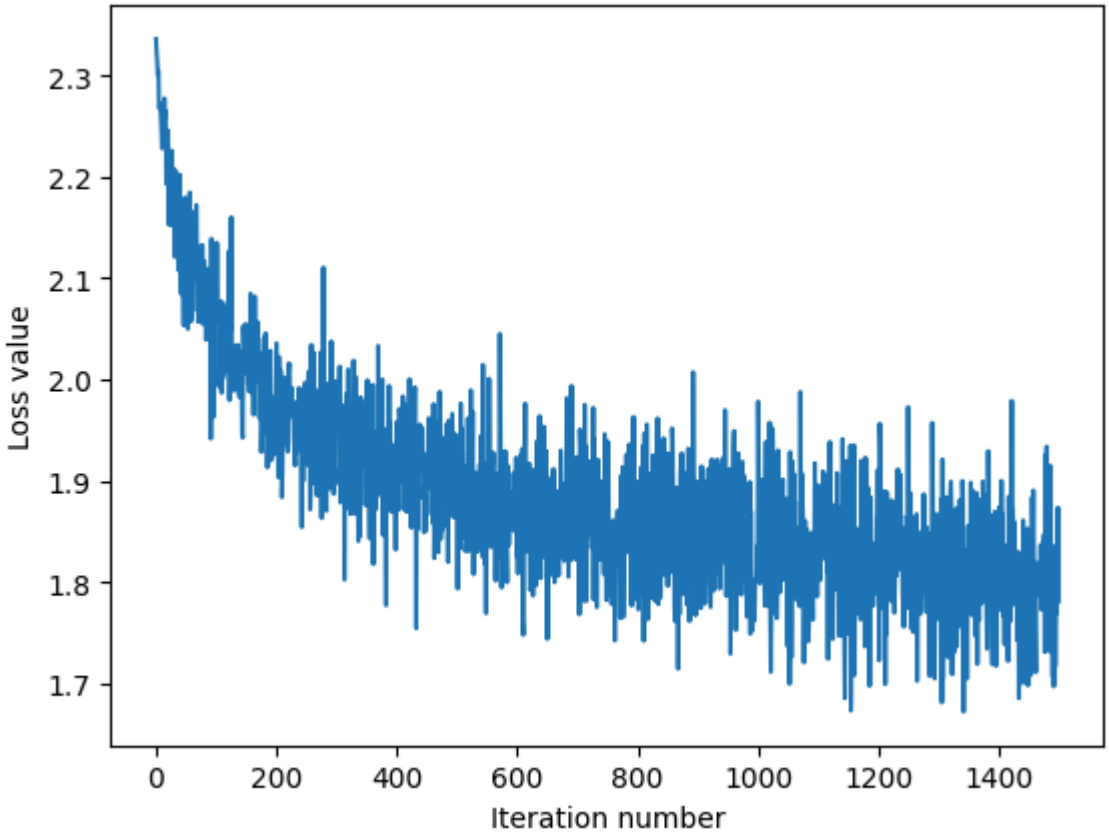
We now implement stochastic gradient descent. This uses the same principles of gradient descent we discussed in class, however, it calculates the gradient by only using examples from a subset of the training set (so each gradient calculation is faster).

In [10]:
```
# Implement softmax.train() by filling in the code to extract a batch of
# and perform the gradient step.
import time


tic = time.time()
loss_hist = softmax.train(X_train, y_train, learning_rate=1e-7,
                      num_iters=1500, verbose=True)
toc = time.time()
print('That took {}s'.format(toc - tic))

plt.plot(loss_hist)
plt.xlabel('Iteration number')
plt.ylabel('Loss value')
plt.show()
```

```
iteration 0 / 1500: loss 2.3365926606637544
iteration 100 / 1500: loss 2.0557222613850827
iteration 200 / 1500: loss 2.0357745120662813
iteration 300 / 1500: loss 1.9813348165609888
iteration 400 / 1500: loss 1.9583142443981612
iteration 500 / 1500: loss 1.8622653073541355
iteration 600 / 1500: loss 1.8532611454359382
iteration 700 / 1500: loss 1.835306222372583
iteration 800 / 1500: loss 1.829389246882764
iteration 900 / 1500: loss 1.8992158530357484
iteration 1000 / 1500: loss 1.97835035402523
iteration 1100 / 1500: loss 1.8470797913532633
iteration 1200 / 1500: loss 1.8411450268664082
iteration 1300 / 1500: loss 1.7910402495792102
iteration 1400 / 1500: loss 1.8705803029382257
That took 6.007515907287598s
```



Evaluate the performance of the trained softmax classifier on the validation data.

In [11]:
```
## Implement softmax.predict() and use it to compute the training and tes

y_train_pred = softmax.predict(X_train)
print('training accuracy: {}'.format(np.mean(np.equal(y_train,y_train_pre
y_val_pred = softmax.predict(X_val)
print('validation accuracy: {}'.format(np.mean(np.equal(y_val, y_val_pred
```

```
training accuracy: 0.3811428571428571
validation accuracy: 0.398
```

Optimize the softmax classifier

In [12]:
```
np.finfo(float).eps
```

Out[12]:
```
2.220446049250313e-16
```

In [14]:
```
# ================================================================ #
# YOUR CODE HERE:
#   Train the Softmax classifier with different learning rates and
#     evaluate on the validation data.
#   Report:
#     - The best learning rate of the ones you tested, determined by best
#     - The best validation accuracy.
#     - Select the model that achieved the best validation accuracy and r
#       its accuracy on the test set.
# ================================================================ #

learning_rates = [1e-8, 5e-8, 1e-7, 5e-7, 1e-6, 5e-6, 1e-5]
best_val = -1
best_softmax = None
best_lr = None

results = {}

for lr in learning_rates:
    # Initialize a new Softmax classifier
    softmax = Softmax(dims=[num_classes, num_features])

    # Train the classifier
    # Using verbose=False to keep the output clean
    loss_hist = softmax.train(X_train, y_train, learning_rate=lr, num_ite

    # Predict on validation set
    y_val_pred = softmax.predict(X_val)
    val_acc = np.mean(np.equal(y_val, y_val_pred))

    results[lr] = val_acc
    print('lr %e val accuracy: %f' % (lr, val_acc))

    # Track the best model
    if val_acc > best_val:
        best_val = val_acc
        best_softmax = softmax
        best_lr = lr

print('\nBest validation accuracy achieved: %f' % best_val)
print('Best learning rate: %e' % best_lr)

# Evaluate on test set with the best model
y_test_pred = best_softmax.predict(X_test)
test_acc = np.mean(np.equal(y_test, y_test_pred))
print('Test accuracy: %f' % test_acc)

# ================================================================ #
# END YOUR CODE HERE
# ================================================================ #
```

```
lr 1.000000e-08 val accuracy: 0.293000
lr 5.000000e-08 val accuracy: 0.366000
lr 1.000000e-07 val accuracy: 0.395000
lr 5.000000e-07 val accuracy: 0.406000
lr 1.000000e-06 val accuracy: 0.413000
lr 5.000000e-06 val accuracy: 0.391000
lr 1.000000e-05 val accuracy: 0.338000

Best validation accuracy achieved: 0.413000
Best learning rate: 1.000000e-06
Test accuracy: 0.404000
```